COMP 3717
Mobile Development with Android Technologies

School	School of Computing and Academic Studies
Program	Computer Systems Technology (CST) Diploma
Course Credits	4
Minimum Passing Grade	50%
Start Date	January 05, 2026
End Date	April 24, 2026
Total Hours*	60
Total Weeks	15
Hours/Weeks	4
Delivery Type	Lecture
Prerequisite(s)	• COMP 2522
CRN	87205

Acknowledgement of Territories

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the Coast Salish Nations of Sḵwx̱wú7mesh [1] (Squamish), səl̓ilwətaʔɬ [2] (Tsleil-Waututh), and xwməθkwəy̓əm [3] (Musqueam).

Instructor Details

Name	Pat McGee
E-mail	Pat_McGee@bcit.ca
Location	
Office Hours	By appointment

Course Description

This hands-on course introduces Android Application Development on mobile and tablet platforms to experienced Java Application Developers. Continuing on from COMP 2526, the Java programming language is used to develop, debug and deploy Android applications. Labs and assignments explore and use the Android SDK tools - avd, adb, ddms, etc., to configure and troubleshoot Android applications on the emulator(s). Topics include: Designing user interfaces with Views, Fragments, Layouts and Activities. Students learn to manage Android Resources, work with Manifest files, Persist data using SQLite DB, Create Services and Content Providers. Other topics of discussion include Location Based Services, Telephony, Multimedia APIs and Cloud to Device Messaging. By the end of this course successful participants will complete a final project which includes building, packaging and publishing an Android application for a mobile device or tablet.

Course Learning Outcomes/Competencies

Upon successful completion of this course, the student will be able to:

- Understand the Android platform and architecture.
- Develop mobile and tablet applications using the Android Application Framework.
- Use the User Interface elements of the Android Framework.
- Create Services for background processing.
- Use and build Content Providers.
- Integrate SQLite Database for Persistence of data.
- Implement Android applications to make use of Location based services.
- Explain how to package and Publish Android applications.
- Access and integrate cloud based services into Android applications.
- Design and build multi-threaded applications for Android devices.
- Develop networked Android applications that connect to web services using HTTP .
- Build Android applications that consume JSON and XML services.

Learning Resources

Required

Android Studio

Optional

You can use your own android device for testing/deploying applications.

Evaluation Criteria

Criteria	%	#	individual/group
Assignment	15	1	individual
Labs	20	9	individual
Lesson exercises	20	9	individual
Midterm	20	1	individual
Final	25	1	individual

Attendance Requirements

Regular attendance in lectures, seminars, and labs is seen as critical to student success, and may be monitored by faculty.

Unapproved absence of 2 or more classes may result in failure or forced withdrawal from the course or program.

Please see Policy 5101 - Student Regulations: https://www.bcit.ca/files/pdf/policies/5101.pdf.

Course Schedule and Assignments

Date	Week	Topics	Deliverables
07-Jan-25	1	System setup and cross platform mobile development introduction	Lesson (due next class)
14-Jan-25	2	State, form inputs, formatting introduction, navigation, life cycle	Lab (due next class) Lesson (due next class)
21-Jan-25	3	Asynchronous tasks, API handling, local DB storage	Lab (due next class) Lesson (due next class)
28-Jan-25	4	Architecture, dialogs, gestures, validation, splash screens	Lab (due next class) Lesson (due next class)
04-Feb-25	5	Native android intro, Jetpack Compose with Kotlin	Lab (due next class) Lesson (due next class)
11-Feb-25	6	Camera, Platform channel	Assignment 1 (due April 14)

18-Feb-25	7	Controls continued, scaffold, list-detail view	Lab (due next class)
25-Feb-25	8	Midterm exam week – non-instructional week	
04-Mar-25	9	Icons, alerts, tabs, app bars, asynchronous tasks, JSON API's	Lab (due next class) Lesson (due next class)
11-Mar-25	10	Room DB, DataStore, UI Continued In-class Instructional Assessments Online	Lab (due next class) Lesson (due next class)
18-Mar-25	11	Reading week	
25-Mar-25	12	Alternative Android libraries	Lab (due next class) Lesson (due next class)
01-Apr-25	13	Alternative Android libraries	Lab (due next class) Lesson (due next class)
08-Apr-25	14	Alternative Android libraries	
15-Apr-25	15	Study / Makeup / Review / Presentation Week	
22-Apr-25	16	Final exam week – non-Instructional week	

** Subject to Change

BCIT Policy

The following statements are in accordance with the BCIT Policies 5101, 4501, 5103, 5104, and 7507, and their accompanying procedures. To review these policies and procedures please click on the links below.

Attendance
In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with their instructor or Program/Department Head, stating the reason for the absence. When absences result in missed safety requirements, exams, or other deadlines, an instructor or Program/Department Head may request appropriate supporting documentation, including a medical note. For all other absences, the student is responsible to ensure they seek out missed information, preferably from a classmate or recording (when available). Please see BCIT Policy 5101 - Student Regulations [4] , and accompanying procedures, for more information.

Attempts
As stated in BCIT Policy 5103 - Student Evaluation [5] , students must successfully complete a course within a maximum of three (3) attempts at the course. Students with two attempts in a single course will be allowed to repeat the course only upon permission from the Associate Dean. Students who have not successfully completed a course within three attempts will not be eligible to graduate from their respective program. For those courses or programs that have Education Council approval, the number of attempts as stated in the evaluation section of the course outline shall apply.

Academic Integrity
It is the responsibility of all students to be familiar with the Student Code of Academic Integrity. Violations of the Code, including plagiarism, cheating, misrepresentation, and academic advantage, are prohibited and will be handled in accordance with BCIT Policy 5104 – Student Code of Academic Integrity [6] , and accompanying procedures.

Accommodation
Any student who may require accommodation from BCIT because of a physical or mental disability should refer to BCIT's Policy 4501 - Accommodation for Students with Disabilities [7] , and contact BCIT's Accessibility Services (SW1 2360, 604-451-6963) at the earliest possible time. Requests for accommodation must be made to Accessibility Services, and should not be made to a course instructor or Program area.

Any student who needs special assistance in the event of a medical emergency or building evacuation (either because of a disability or for any other reason) should promptly inform their course instructor(s) and Accessibility Services of their personal circumstances.

Human Rights, Harassment and Discrimination
The BCIT community is made up of individuals from every ability, background, experience and identity, each contributing uniquely to the richness and diversity of the BCIT community as a whole. In recognition of this, and the intrinsic value of our diversity, BCIT seeks to foster a climate of collaboration, understanding and mutual respect between all members of the community and ensure an inclusive accessible working and learning environment where everyone can succeed. Respect, Diversity, and Inclusion [8] is a supportive resource for both students and employees of BCIT, to foster a respectful learning and working environment. Any student who feels that they are experiencing discrimination or harassment (personal or human rights-related) can confidentially access this resource for advice and support. Please see BCIT Policy 7507 – Harassment and Discrimination [9] and accompanying procedure.

Students should make themselves aware of additional Education, Administration, Safety and other BCIT policies listed at https://www.bcit.ca/about/administration/policies.shtml [10]

Guidelines for School of Computing and Academic Studies

No school specific policies, please refer to main BCIT Policy.

Approved

I verify that the content of this course outline is current.
Pat McGee, Instructor
December 21, 2025

I verify that this course outline has been reviewed.
Chi En Huang, Program Head
January 02, 2026

I verify that this course outline has been reviewed.
Donna Turner, Associate Dean
January 05, 2026

Note: Students will be given reasonable notice if changes are required to the content of this course outline.

*Course hours and credits are calculated per Policy 5012 [11] and the associated procedure [12] .

Total hours – Example of 3 credit lecture/lab course:

- ***Full-time course****: 45 hours of scheduled learning*
- ***Flexible Learning course****: 36 hours of scheduled learning plus 9 hours of independent (non-scheduled, non-instructional) learning*

List of links found on this page

This list includes all links found on this page for your reference.

[1] https://www.squamish.net/
[2] https://twnation.ca/
[3] https://www.musqueam.bc.ca/
[4] https://www.bcit.ca/files/pdf/policies/5101.pdf
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